

Augmenting Collaborative Problem-Solving: Exploring the Design and Use of GenAI for Groupwork

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Abstract

Complex problem-solving and creative work in the real world are rarely individual endeavors and typically unfold within teams and group settings. While advancements in generative artificial intelligence (GenAI) have shown promise in augmenting creativity and productivity, these tools are primarily designed for individual use and overlook group dynamics and the collaborative aspects of teamwork. This workshop will provide a platform for researchers and practitioners to explore the design of future human-AI groups across four key themes: (1) the role of GenAI in group settings, (2) collaborative and multimodal interactions with GenAI, (3) evaluating GenAI's influence within groups and designing for appropriate reliance, and (4) evolving group practices in the presence of GenAI. We hope to build a community and construct alignment across participants around how to pursue research that understands how GenAI can augment, undermine, or bring new practices to collaborative settings and groupwork.

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CCS Concepts

• **Human-centered computing** → **Collaborative and social computing theory, concepts and paradigms; Human computer interaction (HCI).**

Keywords

collaboration, creative teamwork, human-AI teams, generative AI, collective intelligence

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1 Introduction

Generative artificial intelligence (GenAI) technologies are increasingly reshaping how we engage in complex and creative problem-solving tasks [15, 51, 64]. These deep generative models – capable of producing seemingly novel, human-like artifacts – have shown considerable promise in supporting both divergent processes such as brainstorming and ideation [8, 11, 19, 21, 60], and those

that involve convergent thinking like deliberation and decision-making [7, 9, 25, 37, 55]. However, these technologies are underutilized in collaborative settings and are predominantly designed for single user workflows. While this may enhance an individual's productivity, it fundamentally overlooks how problem-solving unfolds within groups acting together [39, 65].

Complex and creative problem-solving tasks are inherently social and collaborative endeavors, and it is often the collaborative aspects of the process that lead to breakthroughs and the ability to tackle complexity in a way that surpasses what an individual might achieve alone [39, 65]. GenAI's evolving ability to perform increasingly complex tasks with minimal human intervention [3, 29, 36], simulate social behavior [34, 42], and dynamically adapt to evolving contextual cues [4, 40, 63] suggests that it could engage in both the analytical and social aspects of problem-solving.

Early explorations of the collaborative use of GenAI [7, 17, 23, 45, 49] along with the emerging capabilities for real-time interactions with conversational [4, 38, 46] and *agentic* or autonomous systems [35, 61] suggests that a future where GenAI can take on independent and context-aware roles within a group might be rapidly approaching [14, 47, 57]. Despite this technological momentum, research on GenAI use in group contexts remains limited, and there are many open questions about its role, efficacy, and ethical considerations in collaborative multi-user settings [26, 48, 52].

It is increasingly evident that GenAI might fundamentally reshape the future of work and transform the dynamic of how people work with technology and with one another [6, 12, 64], with many of its consequences still unfolding. This workshop seeks to begin to address this gap by bringing together researchers and practitioners from across the CSCW and HCI communities to critically examine the emerging role of GenAI in synchronous group interactions and discuss the fundamental socio-technical challenges [1] at the intersection of collaborative use of GenAI and the future of human-AI teamwork. By fostering interdisciplinary dialogue, we hope to develop shared language, surface critical tensions, and identify core research directions for this emerging area. Ultimately, our goal is to lay the groundwork for a research agenda that critically and creatively explores the future of human-AI teamwork and group collaboration – especially for topics of increasing relevance to the CSCW community.

2 Workshop Scope and Goals

Overall, the workshop will focus on the integration of GenAI in **synchronous group settings** and engage participants with four critical themes. Below we briefly describe them and outline the key questions we hope to tackle.

2.1 The Role of GenAI in Group Settings

This theme explores the potential for GenAI to move beyond passive tools and take on more independent and active roles within group settings. More specifically, we will consider what creative and problem-solving processes might benefit most from GenAI contributions, and what roles GenAI can take on to enhance collective intelligence and support effective teamwork. For instance, prior work has explored agents acting as a mimetic agent [35] or devil's advocate [7] during discussions. Teams also tend to become mired

in a narrow set of perspectives [31, 41, 56], and GenAI could be designed introduce diverse viewpoints or help move past a local maxima.

- Which **creative and problem-solving processes** can GenAI support in synchronous group work, and how can it play the role of a **group participant** rather than just a passive tool?
- What **roles** can GenAI play in augmenting group intelligence and enhancing synchronous teamwork?

2.2 Collaborative Interactions with GenAI

How users interact with GenAI is key to effectively harnessing its capabilities [19, 54], and this theme focuses on how the interface design can shape the efficacy of GenAI in group settings. The current trend of predominantly reactive and text-based interfaces that rely on user-generated input, in particular, introduce friction and add an additional overhead that disrupts the flow of collaborative processes [17, 57, 58]. Human collaboration involves a rich tapestry of communication mechanisms and is inherently multimodal, embodied, and situated [10, 18, 22, 30]. Building on early investigations with more expressive technologies like conversational [38] and XR-based embodied interactions [2, 5, 26, 62], we will encourage the exploration of interaction models that traverse text, voice, gestures, and interaction with the environment – modalities that could be critical to integrating GenAI into different group settings [48].

- How can we design GenAI systems that support **synchronous group interaction** during collaborative tasks?
- What **interaction models** best support the varied and evolving roles GenAI can play in group processes?

2.3 Evaluating GenAI's Influence and Designing for Appropriate Reliance

The unintentional consequences of using GenAI for creative and problem-solving processes are beginning to surface [20, 44, 59] as evidence suggests it can lead to over-reliance and reduced critical engagement [24, 33]. While collaborative processes could naturally enhance critical thinking and engagement [13, 16, 66], introducing GenAI into groupwork introduces new complexities [26] that could disrupt group dynamics and power structures. Phenomena like social loafing, matching, or groupthink [31, 32, 41, 56] could also impact how much influence the technology has on a team. This theme will critically engage with how we can evaluate GenAI influence on groups and design systems that ideally amplify human collective abilities without causing deskilling, cognitive disengagement, or undue influence on group dynamics and interpersonal relationships. We hope to discuss how building on theories of group dynamics like the social influence theory [28] could uncover mechanisms to identify detrimental over-reliance within GenAI-augmented groups. We will also reflect on how fostering trust and appropriate reliance in group settings differs from existing interventions for individual users [43, 50, 67]. For instance, how might we allow groups to engage boundary regulation [53] and maintain human agency during collaborative interactions with GenAI?

- How do we **measure success** with respect to the quality of GenAI contributions and evaluate the **influence of GenAI on group** dynamics, processes, and outcomes?

- How do we allow for groups to engage in **collaborative boundary regulation** and dynamically regulate the level of influence machine generated output has on both the process and the outcomes of the group?

2.4 Evolving Group Practices in the Presence of GenAI

Beyond the technological capabilities and system design, the success and impact of GenAI in groupwork will also depend on how workflows and group structures evolve alongside it. This theme aims to guide the design of emerging work practices that optimize GenAI contributions to teamwork and protect the creative aspects of human processes. We will aim to explore how frameworks like collective intelligence [39] and self-organization [27] as well as various group or participatory formats – like roundtable discussions, brainwriting, or division of labor during tasks – can be extended or adapted to support meaningful, safe, and ethical human-AI collaboration in synchronous group settings. Problem-solving also typically occurs over multiple sessions, therefore this theme could also touch on how we might design for the evolution of these collaborative practices over time as group expectations and social norms shift.

- How might GenAI's roles and interaction models reshape **group dynamics** and enable new **creative practices** in groups?
- How can we study and guide the evolution of group processes in the presence of GenAI to develop effective **integration strategies**?

3 Workshop Details and Logistics

3.1 Recruitment and Pre-Workshop Activities

We launched a website at www.augmentedcollaboration.info that contains details about the workshop. We plan on advertising the workshop through social media platforms, public forums, relevant mailing lists, and special interest groups. We will also directly contact researchers and practitioners from both academia and industry whose work aligns with our themes. Finally, we will leverage our local networks and university mailing lists with the hope that the global reach of our organizing team can bring in perspectives from different cultural contexts.

Our call will welcome all who are working on or interested in contributing to an agenda related to the intricacies and challenges of synchronous GenAI use within groups. Participants will be asked to submit a position paper or pictorial¹ (up to 4 pages), and a one-line position statement through an online form. Additional accommodations needed by participants to ensure the workshop is accessible to everyone regardless of their abilities will be solicited as well. A preliminary call for participation is on the website.

All submissions will be reviewed through a single-blind process by at least two organizers. Participants will be selected based on the uniqueness and quality of the content, relevance of their submissions to the workshop goals, and their potential to bring in new perspectives to this community. The overall goal is to ensure a diverse mix of participants with respect to their work, lived

¹<https://dis2019.com/pictorials/>

Time	Activity	Description
09:00-10:00	Introductions and Setting the Stage	Introduction to the workshop agenda, and a creative ice-breaker where participants share their ideas through position statements and symbolic sketches. Organizers then give four short primer talks around the identified themes.
10:00-10:30	Brainstorming AI Roles in Small-Group Rounds	Participants rotate through two themed tables to collaboratively brainstorm creative processes and potential roles for GenAI in groupwork.
10:30-10:45	Break	
10:45-12:30	Co-design Session	Participants co-create low-fidelity prototypes or storyboards for some of the elicited roles and processes, and discuss cross-cutting patterns and insights.
12:30-14:00	Lunch	
14:00-14:45	Eliciting Research Considerations	Participants rotate between co-designed prototypes to note down key design considerations and research questions they feel is relevant to the scenario.
14:45-15:30	Outlining Research Priorities	Participants cluster research questions into themes and use dot voting to prioritize areas of focus and identify topics aligned with their interests.
15:30-15:45	Break	
15:45-17:00	Priority-Based Speed Dating	Participants engage in discussions around their shared interests to spark collaborations and strengthen community connections.
17:00-17:30	Wrap-up and Closing Remarks	Participants reflect their position statement and write a new one. Finally, preferences for sustaining connections and potential collaborative outputs are noted.

Figure 1: Schedule of workshop activities. View detailed descriptions at www.augmentedcollaboration.info/agenda

experiences, and perspectives. Once accepted, we will ask participants to provide camera-ready versions of their submissions in an accessible PDF format (we will offer guidance to make their PDF submissions accessible via our website and provide additional support if requested). These camera-ready versions will be published on the website ahead of the workshop to allow participants the opportunity to familiarize themselves with each other and their perspectives.

4 Workshop Structure and Logistics

The workshop will be facilitated as a full-day in-person event with around 20-30 participants. It will be structured to guide participants through the research themes outlined in Section 2. Before the workshop, the organizers will analyze the position papers submitted by the attendees, group them into 4-5 themes, and assign each to a specific table in the room. We will employ the World Cafe method² to prompt discussion and elicit ideas throughout the workshop. A speed dating exercise will also be used to foster new collaborations around the priorities identified during the workshop. An overview of the agenda is in Table 1 and a full agenda with descriptions for each activity is on the website. We will include alternate means of

²<https://theworldcafe.com/key-concepts-resources/world-cafe-method/>

engagement and accommodate participants' accessibility requests based on best practices³ to the best of our abilities.

No special equipment is needed for the workshop outside of a projector, wifi connection, and tables and chairs. Before the sessions begin, participants will be given a design kit that they will use to participate in the workshop. The design kit will include pens, sticky notes, dot stickers, activity templates, and a booklet with the workshop attendees' details and their submitted positions statements. These materials will be brought to the venue by the organizers.

5 Outcomes and Post-Workshop Plans

Knowledge Dissemination: We will publish a summary of the workshop activities and outcomes on our website. We plan to synthesize the results of the workshop in an ACM *Interactions* or *Communications* article, or as a special issue in an appropriate CSCW/HCI or design-focused journal. Participants who are interested in contributing will be invited to collaborate on this effort, and the final format of the article and publication venues will be collectively decided.

Building a Community: We will establish a community platform (such as a Slack workspace or a Discord server) as a medium that fosters asynchronous dialogue, group collaborations, and networking. Additionally, we will extend an open invitation for non-participants interested in Human-AI groupwork to join the ongoing discourse. If there is sufficient interest, we would also like to host short remote invited talks (from our workshop participants and beyond) every 2-3 months that will be shared on our website and to YouTube. Through these efforts, we aim to sustain momentum beyond the workshop and nurture a vibrant research community around the emerging space of human-AI groups.

6 Organizers

The organizers are all active in research areas that intersect across the design of interactive systems, organizational practices, collaboration, creativity, and collective intelligence. The contact person will be Janet G. Johnson (jgjanet@umich.edu).

Janet G. Johnson is a mixed-methods researcher and Postdoctoral Fellow in the School of Information at the University of Michigan. She received her Ph.D. in Computer Science from the University of California, San Diego. Janet's research focuses on designing and evaluating systems that harness the spatial capabilities of Extended Reality (XR) and the generative capacities of Artificial Intelligence (AI) to support more effective and adaptive collaborative experiences. She also serves as an advisor for the X Reality Safety Intelligence (XRSI), and has served as an AC for CHI and ISMAR.

Steven Rick is a Postdoctoral Research Associate in the Center for Collective Intelligence in the Sloan School of Management at the Massachusetts Institute of Technology. He received his Ph.D. in Computer Science from the University of California, San Diego. As a human-centered design researcher, Steven focuses on how to leverage collective intelligence to design teams of humans and computers that leverage the strengths of each participant to the benefit of the group. Steven has published work in venues such as CHI, IUI, UbiComp, and Collective Intelligence.

Jens Emil Grønbaek is an Assistant Professor at Aarhus University, Denmark. He received his Ph.D. from Aarhus University supported by Microsoft

Research. His current research explores proxemics and flexible interfaces for distributed team collaboration, with the invention of new adaptive interfaces for video conferencing, telepresence, and mixed reality collaboration. Jens Emil has several publications at CHI, UIST, DIS and ISS, and regularly serves as an AC for CHI.

Emily Wong is a Student at The University of Sydney and Research Fellow at The University of Melbourne. Her research focuses on distributed collaboration in mixed reality and she has published work in this area at both CHI and DIS. Beyond her role in academia, Emily has over seven years of industry experience working in human-centred design. Her experience includes designing and facilitating collaborative workshops for government and commercial organisations.

Ming Yin is an Assistant Professor in the Department of Computer Science, Purdue University. She works on examining how to better utilize the wisdom of crowd to enhance machine intelligence, and how to better design intelligent systems that people can understand, trust and engage with effectively. Ming obtained her Ph.D. from Harvard University. Prior to Purdue, Ming spent a year at Microsoft Research New York City as a postdoctoral researcher.

Michael Nebeling is an Associate Professor at the University of Michigan School of Information, where he directs the Information Interaction Lab. His research explores how XR technologies can enhance education, collaboration, and accessibility, with a focus on integrating augmented reality (AR) and artificial intelligence (AI) to improve usability, security, and privacy in user interactions. Prior to Michigan, Michael was a postdoc at the Human-Computer Interaction Institute at Carnegie Mellon University and completed a Ph.D. from ETH Zurich. He also previously spent a sabbatical at Meta's Reality Labs Research.

Mark Klein is an Professor at the School of Collective Intelligence at the University Mohammed VI Polytechnic and Research Scientist with the MIT Center for Collective Intelligence. He develops technology that helps large numbers of people work together more effectively to solve difficult real-world challenges. Mark has a Ph.D. in Artificial Intelligence from the University of Illinois, and has previously held research and teaching positions at Hitachi, Boeing, Pennsylvania State University, the University of Zurich, the Nagoya Institute of Technology, and, for 27 years, as a research scientist with the Center for Collective Intelligence in the Massachusetts Institute of Technology.

Mark Ackerman is the George Herbert Mead Collegiate Professor of Human-Computer Interaction and a Professor in the School of Information and in the Department of Electrical Engineering and Computer Science at the University of Michigan. His major research area is Human-Computer Interaction (HCI), primarily Computer-Supported Cooperative Work (CSCW). He has published widely in HCI and CSCW, investigating expertise sharing (including misinformation), collaborative information access, online knowledge communities, medical settings, and most recently, pervasive environments. Mark is a member of the CHI Academy (HCI Fellow) and an ACM Fellow.

Thomas W. Malone is the Patrick J. McGovern (1959) Professor of Management at the MIT Sloan School of Management and the founding director of the MIT Center for Collective Intelligence. At MIT, he is also a Professor of Information Technology and a Professor of Work and Organizational Studies. Previously, he was the founder and director of the MIT Center for Coordination Science and one of the two founding codirectors of the MIT Initiative on Inventing the Organizations of the 21st Century.

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³For example: <https://coco-net.org/accessibility-guidelines-organizers-facilitators>; <https://www.seedsforchange.org.uk/accessibilitymtg>

References

- [1] Mark S Ackerman. 2000. The intellectual challenge of CSCW: the gap between social requirements and technical feasibility. *Human-Computer Interaction* 15, 2-3 (2000), 179–203.
- [2] Mark Billingham, Pablo Cesar, Mar Gonzalez-Franco, Katherine Isbister, Julie Williamson, and Alexandra Kitson. 2024. Social XR: The Future of Communication and Collaboration (Dagstuhl Seminar 23482). (2024).
- [3] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. 2023. Autonomous chemical research with large language models. *Nature* 624, 7992 (2023), 570–578.
- [4] Riccardo Bovo, Steven Abreu, Karan Ahuja, Eric J Gonzalez, Li-Te Cheng, and Mar Gonzalez-Franco. 2024. Embardiment: an embodied ai agent for productivity in xr. *arXiv preprint arXiv:2408.08158* (2024).
- [5] Efe Bozkir, Süleyman Özdel, Ka Hei Carrie Lau, Mengdi Wang, Hong Gao, and Enkelejda Kasneci. 2024. Embedding large language models into extended reality: Opportunities and challenges for inclusion, engagement, and privacy. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces*. 1–7.
- [6] Mauro Cazzaniga, Ms Florence Jaumotte, Longji Li, Mr Giovanni Melina, Augustus J Panton, Carlo Pizzinelli, Emma J Rockall, and Ms Marina Mendes Tavares. 2024. *Gen-AI: Artificial intelligence and the future of work*. International Monetary Fund.
- [7] Chun-Wei Chiang, Zhuoran Lu, Zhuoyan Li, and Ming Yin. 2024. Enhancing AI-Assisted Group Decision Making through LLM-Powered Devil’s Advocate. In *Proceedings of the 29th International Conference on Intelligent User Interfaces*. 103–119.
- [8] Li-Yuan Chiou, Peng-Kai Hung, Rung-Huei Liang, and Chun-Teng Wang. 2023. Designing with AI: an exploration of co-ideation with image generators. In *Proceedings of the 2023 ACM designing interactive systems conference*. 1941–1954.
- [9] Zijian Ding, Arvind Srinivasan, Stephen MacNeil, and Joel Chan. 2023. Fluid transformers and creative analogies: Exploring large language models’ capacity for augmenting cross-domain analogical creativity. In *Proceedings of the 15th Conference on Creativity and Cognition*. 489–505.
- [10] Gwyneth Doherty-Sneddon, Anne Anderson, Claire O’malley, Steve Langton, Simon Garrod, and Vicki Bruce. 1997. Face-to-face and video-mediated communication: A comparison of dialogue structure and task performance. *Journal of experimental psychology: applied* 3, 2 (1997), 105.
- [11] Anil R Doshi and Oliver P Hauser. 2024. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science Advances* 10, 28 (2024), eadn5290.
- [12] Tyna Eloundou, Sam Manning, Pamela Mishkin, and Daniel Rock. 2024. GPTs are GPTs: Labor market impact potential of LLMs. *Science* 384, 6702 (2024), 1306–1308.
- [13] Stephen M Fiore, Michael A Rosen, Kimberly A Smith-Jentsch, Eduardo Salas, Michael Letsky, and Norman Warner. 2010. Toward an understanding of macrocognition in teams: Predicting processes in complex collaborative contexts. *Human factors* 52, 2 (2010), 203–224.
- [14] Christopher Flathmann, Beau Schellble, Brock Tubre, Nathan McNeese, and Paige Rodeghero. 2020. Invoking principles of groupware to develop and evaluate present and future human-agent teams. In *Proceedings of the 8th International Conference on Human-Agent Interaction*. 15–24.
- [15] Fiona Fui-Hoon Nah, Ruilin Zheng, Jingyuan Cai, Keng Siau, and Langtao Chen. 2023. Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration. , 277–304 pages.
- [16] Vera Hagemann and Annette Kluge. 2017. Complex problem solving in teams: the impact of collective orientation on team process demands. *Frontiers in psychology* 8 (2017), 1730.
- [17] Yuanning Han, Ziyi Qiu, Jiale Cheng, and RAY LC. 2024. When Teams Embrace AI: Human Collaboration Strategies in Generative Prompting in a Creative Design Task. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–14.
- [18] Steve Harrison and Paul Dourish. 1996. Re-place-ing space: the roles of place and space in collaborative systems. In *Proceedings of the 1996 ACM conference on Computer supported cooperative work*. 67–76.
- [19] Jennifer L Heyman, Steven R Rick, Gianni Giacomelli, Haoran Wen, Robert Laubacher, Nancy Taubenslag, Max Knicker, Younes Jeddi, Pranav Ragupathy, Jared Curhan, et al. 2024. Supermind Ideator: How Scaffolding Human-AI Collaboration Can Increase Creativity. In *Proceedings of the ACM Collective Intelligence Conference*. 18–28.
- [20] Wayne Holmes. 2023. The unintended consequences of artificial intelligence and education. (2023).
- [21] Kent F Hubert, Kim N Awa, and Darya L Zabelina. 2024. The current state of artificial intelligence generative language models is more creative than humans on divergent thinking tasks. *Scientific Reports* 14, 1 (2024), 3440.
- [22] John A Hughes, Wolfgang Prinz, Tom Rodden, Kjeld Schmidt, and Toni Robertson. 1997. Cooperative work and lived cognition: a taxonomy of embodied actions. In *Proceedings of the Fifth European Conference on Computer Supported Cooperative Work*. Springer, 205–220.
- [23] Shota Imamura, Hirotaka Hiraki, and Jun Rekimoto. 2024. Serendipity Wall: A Discussion Support System Using Real-time Speech Recognition and Large Language Model. In *Proceedings of the Augmented Humans International Conference 2024*. 237–247.
- [24] Maia Jacobs, Melanie F Pradier, Thomas H McCoy Jr, Roy H Perlis, Finale Doshi-Velez, and Krzysztof Z Gajos. 2021. How machine-learning recommendations influence clinician treatment selections: the example of antidepressant selection. *Translational psychiatry* 11, 1 (2021), 108.
- [25] Youngseung Jeon, Seungwan Jin, Patrick C Shih, and Kyungsik Han. 2021. FashionQ: an ai-driven creativity support tool for facilitating ideation in fashion design. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–18.
- [26] Janet G Johnson, Macarena Peralta, Mansanjam Kaur, Ruijie Sophia Huang, Sheng Zhao, Ruijia Guan, Shwetha Rajaram, and Michael Nebeling. 2025. Exploring Collaborative GenAI Agents in Synchronous Group Settings: Eliciting Team Perceptions and Design Considerations for the Future of Work. *arXiv preprint arXiv:2504.14779* (2025).
- [27] Simon Kaplan and Lesley Seebeck. 2001. Harnessing complexity in CSCW. In *ECSCW 2001: Proceedings of the Seventh European Conference on Computer Supported Cooperative Work 16–20 September 2001, Bonn, Germany*. Springer, 359–378.
- [28] Herbert C Kelman. 2006. Interests, relationships, identities: Three central issues for individuals and groups in negotiating their social environment. *Annu. Rev. Psychol.* 57, 1 (2006), 1–26.
- [29] Geunwoo Kim, Pierre Baldi, and Stephen McAleer. 2023. Language models can solve computer tasks. *Advances in Neural Information Processing Systems* 36 (2023), 39648–39677.
- [30] David Kirsh. 2013. Embodied cognition and the magical future of interaction design. *ACM Transactions on Computer-Human Interaction (TOCHI)* 20, 1 (2013), 1–30.
- [31] Mark Klein. 2011. How to harvest collective wisdom on complex problems: An introduction to the mit deliberatorium. *Center for Collective Intelligence working paper* 79 (2011).
- [32] Nicholas W Kohn and Steven M Smith. 2011. Collaborative fixation: Effects of others’ ideas on brainstorming. *Applied Cognitive Psychology* 25, 3 (2011), 359–371.
- [33] Hao-Ping Hank Lee, Advait Sarkar, Lev Tankelevitch, Ian Drosos, Sean Rintel, Richard Banks, and Nicholas Wilson. 2025. The Impact of Generative AI on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confidence Effects From a Survey of Knowledge Workers. (2025).
- [34] Yan Leng and Yuan Yuan. 2023. Do LLM Agents Exhibit Social Behavior? *arXiv preprint arXiv:2312.15198* (2023).
- [35] Joanne Leong, John Tang, Edward Cutrell, Sasa Junuzovic, Gregory Paul Baribault, and Kori Inkpen. 2024. Dittos: Personalized, Embodied Agents That Participate in Meetings When You Are Unavailable. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–28.
- [36] Vadim Liventsev, Anastasiia Grishina, Aki Härmä, and Leon Moonen. 2023. Fully autonomous programming with large language models. In *Proceedings of the Genetic and Evolutionary Computation Conference*. 1146–1155.
- [37] Shuai Ma, Qiaoyi Chen, Xinru Wang, Chengbo Zheng, Zhenhui Peng, Ming Yin, and Xiaojuan Ma. 2024. Towards human-ai deliberation: Design and evaluation of llm-empowered deliberative ai for ai-assisted decision-making. *arXiv preprint arXiv:2403.16812* (2024).
- [38] Amama Mahmood, Junxiang Wang, Bingsheng Yao, Dakuo Wang, and Chien-Ming Huang. 2023. LLM-Powered Conversational Voice Assistants: Interaction Patterns, Opportunities, Challenges, and Design Guidelines. *arXiv preprint arXiv:2309.13879* (2023).
- [39] Thomas W Malone. 2018. *Superminds: The surprising power of people and computers thinking together*. Little, Brown Spark.
- [40] Michael Muller, Anna Kantosalo, Mary Lou Maher, Charles Patrick Martin, and Greg Walsh. 2024. GenAICHI 2024: Generative AI and HCI at CHI 2024. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [41] Michael Pacanowsky. 1995. Team tools for wicked problems. *Organizational dynamics* 23, 3 (1995), 36–51.
- [42] Joon Sung Park, Lindsay Popowski, Carrie Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2022. Social simulacra: Creating populated prototypes for social computing systems. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. 1–18.
- [43] Charvi Rastogi, Yunfeng Zhang, Dennis Wei, Kush R Varshney, Amit Dhurandhar, and Richard Tomsett. 2022. Deciding fast and slow: The role of cognitive biases in ai-assisted decision-making. *Proceedings of the ACM on Human-computer Interaction* 6, CSCW1 (2022), 1–22.
- [44] Roberto Redaelli. 2024. Intentionality gap and preter-intentionality in generative artificial intelligence. *AI & SOCIETY* (2024), 1–8.
- [45] Mohammad Amin Samadi, Spencer JaQuay, Jing Gu, and Nia Nixon. 2024. The AI Collaborator: Bridging Human-AI Interaction in Educational and Professional Settings. *arXiv preprint arXiv:2405.10460* (2024).

- [46] Katie Seaborn, Norihisa P Miyake, Peter Pennefather, and Mihoko Otake-Matsuura. 2021. Voice in human-agent interaction: A survey. *ACM Computing Surveys (CSUR)* 54, 4 (2021), 1–43.
- [47] Isabella Seeber, Eva Bittner, Robert O Briggs, Triparna De Vreede, Gert-Jan De Vreede, Aaron Elkins, Ronald Maier, Alexander B Merz, Sarah Oeste-Reiß, Nils Randrup, et al. 2020. Machines as teammates: A research agenda on AI in team collaboration. *Information & management* 57, 2 (2020), 103174.
- [48] William Seymour and Emilee Rader. 2024. Speculating About Multi-user Conversational Interfaces and LLMs: What If Chatting Wasn't So Lonely?. In *Proceedings of the 6th ACM Conference on Conversational User Interfaces*. 1–4.
- [49] Orit Shaer, Angelora Cooper, Osnat Mokryn, Andrew L Kun, and Hagit Ben Shoshan. 2024. AI-Augmented Brainwriting: Investigating the use of LLMs in group ideation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [50] Hua Shen, Tiffany Kneare, Reshmi Ghosh, Kenan Alkiek, Kundan Krishna, Yachuan Liu, Ziqiao Ma, Savvas Petridis, Yi-Hao Peng, Li Qiwei, et al. 2024. Towards bidirectional human-ai alignment: A systematic review for clarifications, framework, and future directions. *arXiv preprint arXiv:2406.09264* (2024).
- [51] Yang Shi, Tian Gao, Xiaohan Jiao, and Nan Cao. 2023. Understanding design collaboration between designers and artificial intelligence: A systematic literature review. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW2 (2023), 1–35.
- [52] Dominik Siemon. 2022. Elaborating team roles for artificial intelligence-based teammates in human-AI collaboration. *Group Decision and Negotiation* 31, 5 (2022), 871–912.
- [53] Frederic Stutzman and Woodrow Hartzog. 2012. Boundary regulation in social media. In *Proceedings of the ACM 2012 conference on computer supported cooperative work*. 769–778.
- [54] Hari Subramonyam, Roy Pea, Christopher Pondoc, Maneesh Agrawala, and Colleen Seifert. 2024. Bridging the Gulf of Envisioning: Cognitive Challenges in Prompt Based Interactions with LLMs. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–19.
- [55] Tommaso Turchi, Silvio Carta, Luciano Ambrosini, and Alessio Malizia. 2023. Human-AI co-creation: evaluating the impact of large-scale text-to-image generative models on the creative process. In *International Symposium on End User Development*. Springer, 35–51.
- [56] Amos Tversky and Daniel Kahneman. 1974. Judgment under Uncertainty: Heuristics and Biases: Biases in judgments reveal some heuristics of thinking under uncertainty. *science* 185, 4157 (1974), 1124–1131.
- [57] Simone van den Broek, Supraja Sankaran, Jan de Wit, and Alwin de Rooij. 2024. Exploring the supportive role of artificial intelligence in participatory design: a systematic review. In *Proceedings of the Participatory Design Conference 2024: Exploratory Papers and Workshops-Volume 2*. 37–44.
- [58] Mathias Peter Verheijden and Mathias Funk. 2023. Collaborative diffusion: Boosting designerly co-creation with generative AI. In *Extended abstracts of the 2023 CHI conference on human factors in computing systems*. 1–8.
- [59] Krzysztof Wach, Cong Doanh Duong, Joanna Ejdy, Rūta Kazlauskaitė, Paweł Korzyński, Grzegorz Mazurek, Joanna Paliszkievicz, and Ewa Ziemia. 2023. The dark side of generative artificial intelligence: A critical analysis of controversies and risks of ChatGPT. *Entrepreneurial Business and Economics Review* 11, 2 (2023), 7–30.
- [60] Samangi Wadinambiarachchi, Ryan M Kelly, Saumya Pareek, Qiushi Zhou, and Eduardo Velloso. 2024. The Effects of Generative AI on Design Fixation and Divergent Thinking. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–18.
- [61] Hongyu Wan, Jinda Zhang, Abdulaziz Arif Suria, Bingsheng Yao, Dakuo Wang, Yvonne Coady, and Mirjana Prpa. 2024. Building LLM-based AI Agents in Social Virtual Reality. In *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 65, 7 pages. <https://doi.org/10.1145/3613905.3651026>
- [62] Hongyu Wan, Jinda Zhang, Abdulaziz Arif Suria, Bingsheng Yao, Dakuo Wang, Yvonne Coady, and Mirjana Prpa. 2024. Building LLM-based AI Agents in Social Virtual Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [63] Yidong Wang, Qi Guo, Wenjin Yao, Hongbo Zhang, Xin Zhang, Zhen Wu, Meishan Zhang, Xinyu Dai, Qingsong Wen, Wei Ye, et al. 2024. Autosurvey: Large language models can automatically write surveys. *Advances in Neural Information Processing Systems* 37 (2024), 115119–115145.
- [64] Allison Woodruff, Renee Shelby, Patrick Gage Kelley, Steven Rousso-Schindler, Jamila Smith-Loud, and Lauren Wilcox. 2024. How knowledge workers think generative ai will (not) transform their industries. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–26.
- [65] Stefan Wuchty, Benjamin F Jones, and Brian Uzzi. 2007. The increasing dominance of teams in production of knowledge. *Science* 316, 5827 (2007), 1036–1039.
- [66] Enwei Xu, Wei Wang, and Qingxia Wang. 2023. The effectiveness of collaborative problem solving in promoting students' critical thinking: A meta-analysis based on empirical literature. *Humanities and Social Sciences Communications* 10, 1 (2023), 1–11.
- [67] ShunYi Yeo, Gionnieve Lim, Jie Gao, Weiyu Zhang, and Simon Tangi Perrault. 2024. Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–32.